Topological Memory.

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Introduction

Today:

- Proof.
    1. Walktrough the math

Our Code

Code

In this session, we consider LDPC codes, where each bit adjoins to Δ_1 checks, and each check adjoins to Δ_2 bits. We denote by $\Delta = \Delta_1\Delta_2$.

Local Decoder

definition

We say that D is a local decoder if when applied against $2p(1-p)$-depolarized channel:

- There exists function $M : (0,1) \to (0,1)$ such that the probability of each bit being flipped is $M(2p(1-p)) < p^{\frac{100}{99}}$.
- There is a constant l such that for any bit, there are only $(\Delta)^l$ bits on which the event of it being flipped depends.

For example, if we take k large enough and consider the (n,k)-Toric, then both the swift-rule and the flip upon majority are local decoders.

Storing Logical State - Classical Statement

Logical State

For a random variable X we denote by:

1. $\mathbf{E}_{\sim p}[X]$ - its expectation when measured after a single application of the p-depolarizing channel.
2. $\mathbf{E}_{\sim q^{(t)}}[X]$ - its expectation when measured after t rounds, each initiated by a p-depolarized channel followed by an application of local decoder D.

The Theorem - Classical Statement

Theorem

There is $p_{\text{th}} \in (0,1)$ such that for any $p < p_{\text{th}}$ and any bounded function f with a supremum $|f|$ it holds:

$$\mathbf{E}_{\sim q^{(t)}}[f] \leq \mathbf{E}_{\sim p}[f] + t|f|e^{-\Theta(\sqrt{n})}$$

Lemma

There exists a constant $c > 0$, such:

$$\mathbf{E}_{\sim 2p(1-p)}[f \circ D] \leq \mathbf{E}_{\sim p}[f] + |f|e^{-c\sqrt{n}}$$

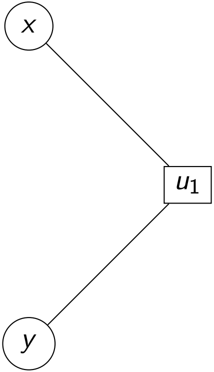
Theorem Proof

By induction on t.

$$\begin{aligned}
\mathbf{E}_{\sim q^{(t)}}[f] &= \mathbf{E}_{\sim q^{(t-1)}}\left[\mathbf{E}_{\sim p}[f \circ D]\right] \\
&\leq \mathbf{E}_{\sim p}\left[\mathbf{E}_{\sim p}[f \circ D]\right] + (t-1)|\mathbf{E}_{p}[f \circ D]|e^{-c\sqrt{n}} \\
&\leq \mathbf{E}_{\sim 2p(1-p)}[f \circ D] + (t-1)|f|e^{-c\sqrt{n}} \\
&\leq \mathbf{E}_{\sim p}[f] + t|f|e^{-c\sqrt{n}}
\end{aligned}$$

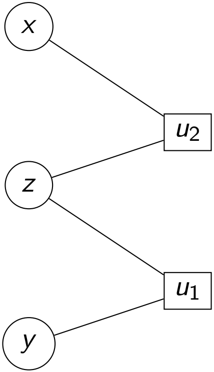
When in the last transition we used the lemma.

Dependence Type I



$$
\begin{aligned}
&\mathbf{Pr}\left[(x, y) \in e \middle| u_1 \text{ is satisfied}\right] \\
&= \mathbf{Pr}\left[x \in e \middle| u_1 \text{ is satisfied}\right] \mathbf{Pr}\left[y \in e \middle| u_1 \text{ is satisfied}\right] \\
&\leq \mathbf{Pr}\left[x \in e\right] \mathbf{Pr}\left[y \in e\right]
\end{aligned}
$$

Dependence Type II



$$\begin{aligned}&\mathbf{Pr}\left[(x, y) \in e | z \text{ is not flipped}\right]\\ &= \mathbf{Pr}\left[x \in e | z \text{ is not flipped}\right] \mathbf{Pr}\left[y \in e | z \text{ is not flipped}\right]\\ &\leq \mathbf{Pr}\left[x \in e\right] \mathbf{Pr}\left[y \in e\right]\end{aligned}$$

Definition

Consider the moment after the application of the depolarizing channel, and right before the correction by the decoder.
For a subset of qubits S, we say that a qubit $x \in S$ is depended if there exists $y \in S$ such:

1. Either x and y share a check that is not satisfied. (Type I)
2. Or, x and y have checks which share a bit that was flipped by the application of the channel. (Type II)

Dependence

Claim

Consider the application of a $2p(1-p)$-depolarizing channel. Let S be a subset of qubits chosen uniformly random [1], then the probability that there are more than $\frac{1}{100}$ depended qubits in S is less than $2e^{-c\sqrt{n}}$.

[1] each qubit is taken with probability $\frac{1}{2}$

Dependence

Proof

First, for an arbitrary S, denote by J the qubits that depend on qubits in S. Then, the expected number of depended qubits in S is:

$$\mathbf{E}_{\sim 2p(1-p)}\left[\text{Type II depended qubits in} S\right] \\ \leq \mathbf{E}_{\sim 2p(1-p)}\left[|J|\right] \leq 2p(1-p)\Delta^{l}|S|$$

Similarly, let $J\prime$ be the qubits in S that connected to unsatisfied check, then:

$$\mathbf{E}_{\sim 2p(1-p)}\left[\text{Type I depended qubits in} S\right] \\ \leq \mathbf{E}_{\sim 2p(1-p)}\left[|J|'\right] \leq \frac{1-(1-2p)^{\Delta_2}}{2}\Delta^{l}|S|$$

Dependence

We are going to use the following inequality:

Theorem 1 (McDiarmid's inequality). *Let $X_1, \ldots, X_n$ be independent random variables, where X_i has range $\mathcal{X}_i$. Let $f\colon \mathcal{X}_1 \times \cdots \times \mathcal{X}_n \to \mathbb{R}$ be any function with the $(c_1, \ldots, c_n)$*-bounded differences property*: for every $i = 1, \ldots, n$ and every $(x_1, \ldots, x_n), (x'_1, \ldots, x'_n) \in \mathcal{X}_1 \times \cdots \times \mathcal{X}_n$ that differ only in the i-th corodinate ($x_j = x'_j$ for all $j \neq i$), we have*

$$|f(x_1, \ldots, x_n) - f(x'_1, \ldots, x'_n)| \leq c_i.$$

For any $t > 0$,

$$\Pr(f(X_1, \ldots, X_n) - \mathbb{E}[f(X_1, \ldots, X_n)] \geq t) \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^n c_i^2}\right).$$

Dependence

Thus, the probability that the number of depended qubits in S is $|S|^{\frac{3}{4}}$ far from the expectation:

$$\mathbf{Pr}_{\sim 2p(1-p)}\left[\# - \mathbf{E}\left[\#\right] \geq |S|^{\frac{3}{4}}\right] \leq e^{-2\frac{|S|^{\frac{3}{2}}}{n\Delta^{l}}}$$

So, for any $|S| > \alpha n$ we have:

$$\mathbf{Pr}_{\sim 2p(1-p)}\left[\# - \mathbf{E}\left[\#\right] \geq |S|^{\frac{3}{4}}\right] \leq e^{-\frac{2\alpha^{\frac{2}{3}}}{\Delta^{l}}\sqrt{n}}$$

Dependence

Thus for small enough $p \in (0,1)$, with probability greater than $1 - 2e^{-\frac{2\alpha^{\frac{2}{3}}}{\Delta^l}\sqrt{n}}$, the number of depended bits is less than:

$$\left(2p(1-p) + \frac{1-(1-2p)^{\Delta_2}}{2}\right)\Delta^l|S| + |S|^{\frac{3}{4}} \to \frac{|S|}{100}$$

Lemma Proof

Denote by S the subset of the flipped qubits, and denote by $de(S)$ the number of depended qubits in S.

$$\begin{aligned}
&\mathbf{E}_{\sim 2p(1-p)}\left[f \circ D\right] = \sum f \circ D(S) \cdot \mathbf{Pr}\left[S\right] \\
&\leq \sum_{\substack{de(S)<\frac{1}{100}|S| \\ |S|\geq \alpha n}} f \circ D(S) \cdot \mathbf{Pr}\left[S\right] + 2|f|e^{-c\sqrt{n}} \\
&\leq \sum_{\substack{de(S)<\frac{1}{100}|S| \\ |S|\geq \alpha n}} f(S)M(2p(1-p))^{\frac{99}{100}|S|} + 2|f|e^{-c\sqrt{n}}
\end{aligned}$$

Lemma Proof

$$
\begin{aligned}
&\leq \sum_{\substack{de(S) < \frac{1}{100}|S| \\ |S| \geq \alpha n}} f(S) p^{|S|} + 2|f| e^{-c\sqrt{n}} \\
&\leq \sum_{S} f(S) p^{|S|} + 2|f| e^{-c\sqrt{n}} \\
&\leq \mathbf{E}_{\sim p}\left[f(S) \right] + 2|f| e^{-c\sqrt{n}}
\end{aligned}
$$

Storing Logical State - Quantum Statement

Logical State

Let $\rho^{(0)}$ be a mixed state over codewords. Define:

$$\rho^{(t)} = D \circ \mathcal{N} \rho^{(t-1)}$$

1. $\mathbf{E}_{\sim p}[X]$ - its expectation when measured after a single application of the p-depolarizing channel.
2. $\mathbf{E}_{\sim q^{(t)}}[X]$ - its expectation when measured after t rounds, each initiated by a p-depolarized channel followed by an application of local decoder D.

The Theorem - Quantum Statement

Theorem

There is $p_{\text{th}} \in (0,1)$ such that for any $p < p_{\text{th}}$ and any bounded Hermitian operator X we have:

$$\textbf{Tr } X\rho^{(t)} \leq \textbf{ Tr } X\mathcal{N}_p(\rho) + t||X||e^{-\Theta(\sqrt{n})}$$

Lemma

There exists a constant $c > 0$, such:

$$\textbf{Tr } XD \circ \mathcal{N}_p \circ \mathcal{N}_p(\rho) \leq \textbf{ Tr } X\mathcal{N}_p(\rho) + 2||X||e^{-\Theta(\sqrt{n})}$$

Theorem Proof

By induction on t.

$$\begin{aligned}\textbf{Tr } X\rho^{(t)} &= \textbf{Tr } XD \circ \mathcal{N}_p \rho^{(t-1)} \\ &\leq \textbf{Tr } XD \circ \mathcal{N}_p \circ \mathcal{N}_p(\rho) + (t-1)||X||e^{-\Theta(\sqrt{n})} \\ &\leq \textbf{Tr } X\mathcal{N}_p(\rho) + t||X||e^{-\Theta(\sqrt{n})}\end{aligned}$$

When in the last transition we used the lemma.

Corollary

$$\mathbf{Tr}\ (I - X)\,\rho^{(t)} \leq\ \mathbf{Tr}\ (I - X)\,\mathcal{N}_p\,(\rho) + t||I - X||e^{-\Theta(\sqrt{n})}$$
$$\Leftrightarrow \mathbf{Tr}\ X\rho^{(t)} \geq\ \mathbf{Tr}\ X\mathcal{N}_p\,(\rho) - t||I - X||e^{-\Theta(\sqrt{n})}$$